

Report
Of
CYBERSECURITY AND VULNERABILITY ASSESSMENT
(internship Report)

Submitted
To
**School of Computer Science and Engineering
IILM University, Greater Noida, U.P.**

In partial fulfilment of the requirement of degree of
B.Tech CSE



By
Mukesh Bhandari
CS-2341752
Batch:2026-27

CANDIDATE'S DECLARATION

I, Mukesh Bhandari do hereby declare that the internship report titled “**Cybersecurity and Vulnerability Assessment**” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

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Roll No:CS-2341752

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to give a special mention to my colleagues. Last but not least I am grateful to all the faculty members of the Academy of Skill Development for their support.

I also extend my gratitude to my faculty at IILM University, School of Computer Science and Engineering, for their continuous guidance and support throughout this internship and report-writing process.

Signature

Student Name: Mukesh Bhandari

Roll No:CS-2341752

INTERNSHIP COMPLETION CERTIFICATE

Academy of Skill Development
A non-profit trust registered with Govt. of West Bengal U/S 60 and Rule 69 with registration number -190307248
Industrial Internship Certificate

This certificate is awarded to

MUKESH BHANDARI

of

IILM UNIVERSITY, GREATER NOIDA



for successfully completing the **Industrial Internship** on

CYBER SECURITY AND ETHICAL HACKING

from

August 1, 2026 to September 12, 2026 (4 Weeks)

and implementing the project titled

PASSWORD STRENGTH CHECKER





Certificate ID: **ASD/CYB/IIL/PAS/105292**
Issue Date: **September 12, 2026**



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CHAPTER 4: PROJECT DESCRIPTION

4.1 Introduction

This report presents an account of the internship undertaken by me, Mukesh Bhandari, in the field of Cybersecurity and Vulnerability Assessment at the Academy of Skill Development, during August–September 2026. The internship provided an opportunity to gain practical exposure to cybersecurity concepts and tools, allowing me to apply theoretical knowledge through hands-on activities in controlled laboratory environments.

The training introduced me to fundamental concepts of network security, vulnerability assessment, ethical hacking, and system security, and gave me first-hand experience with the workflow a security analyst follows: setting up an isolated lab, scanning a target for open services, and researching the vulnerabilities those services may expose. Overall, the internship strengthened my practical understanding of cybersecurity and built hands-on familiarity with commonly used security tools and vulnerability-assessment practices.

4.2 Organization Profile

Academy of Skill Development is a training and skill-development organization that provides practical learning in technical and professional fields. During my internship, the organization provided training in cybersecurity, ethical hacking, vulnerability assessment, and related security practices, structured around hands-on activities in controlled laboratory environments.

4.3 Problem Statement

Organizations increasingly need personnel who can identify and assess security weaknesses in their systems before those weaknesses are exploited, yet this kind of applied, tool-based experience is difficult to gain from classroom study alone. This internship was designed to close that gap by giving trainees direct, supervised practice with the reconnaissance and assessment techniques used in real vulnerability assessments — without the legal and safety risks of testing live, non-consenting

systems. To this end, intentionally vulnerable systems (Metasploitable 2 and Windows 7) were used as safe stand-ins for real-world targets.

4.4 Project Objectives

The main objectives of this internship were:

- To understand the fundamentals of cybersecurity and ethical hacking.
- To learn and use Kali Linux and basic Linux commands.
- To perform network scanning and service identification using Nmap.
- To understand and assess vulnerabilities in Metasploitable 2 and Windows 7 in controlled environments.
- To gain practical knowledge of VirusTotal, Exploit Database, and Shodan.
- To develop practical skills in vulnerability assessment and security analysis.
- To understand the importance of ethical and authorized security testing.

4.5 Scope of the Project

The internship was limited to educational, authorized, and fully controlled environments — no external, production, or third-party systems were scanned or tested at any point. Within that boundary, the work covered lab setup and Linux fundamentals, service discovery with Nmap, and vulnerability study on two designated lab targets (Metasploitable 2 and Windows 7).

The scope did not extend to exploitation or post-exploitation activity against production systems, penetration testing engagements with a client, or the development of original exploit code; the focus throughout stayed on identification, understanding, and reporting of vulnerabilities rather than active compromise.

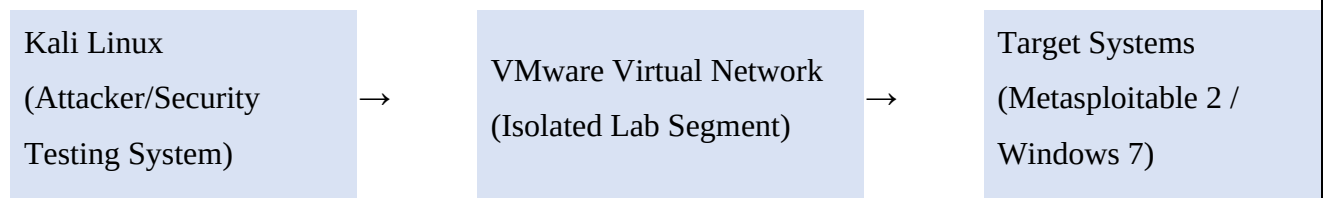
4.6 Technologies and Tools Used

- Kali Linux – Cybersecurity and security testing.
- Nmap – Network and port scanning.
- Metasploitable 2 & Windows 7 – Vulnerability assessment in a lab.

- VMware – Virtual lab environment.
- VirusTotal – Security analysis.
- Exploit Database – Vulnerability research.
- Shodan – Information about internet-connected devices.

4.7 System Architecture

The internship laboratory was designed as a virtualized, isolated cybersecurity environment. Kali Linux served as the primary security-testing system, while Metasploitable 2 and Windows 7 acted as controlled target systems reachable only within the virtual lab network. Nmap was used from the Kali Linux system to scan and identify ports and services running on the targets.



Keeping the target systems on an isolated virtual network (rather than a shared or internet-facing one) was central to the design: it meant scanning and vulnerability-study activity stayed fully contained, with no possibility of affecting systems outside the lab.

4.8 Methodology

- The internship followed a practical, step-by-step approach:
- Lab Setup: Set up Kali Linux and virtual machines using VMware.
- Linux Practice: Learned basic Linux commands and system administration.
- Network Scanning: Used Nmap to identify open ports and running services.
- Vulnerability Assessment: Studied vulnerabilities in Metasploitable 2 and Windows 7 in controlled environments.
- Security Research: Used VirusTotal, Exploit Database, and Shodan to study security-related information.

4.9 Expected Outcomes

- The expected outcomes of the internship were:
- Develop a basic understanding of cybersecurity and ethical hacking.
- Gain practical knowledge of Kali Linux and Nmap.
- Understand network scanning and vulnerability assessment.
- Learn to identify and analyze common security vulnerabilities in controlled environments.
- Gain familiarity with VirusTotal, Exploit Database, and Shodan.
- Develop practical skills for further learning and a career in cybersecurity..

4.10 Certificates of Completion and Communication Proof

The Internship Certificate issued by the Academy of Skill Development is attached in this section as proof of the internship undertaken and successfully completed.

4.11 Client Requirement Handling and Communication Flow

Task requirements and instructions during the internship were communicated by the lab supervisor at the start of each module, typically as a short written brief outlining the target system, the tool to be used, and the specific objective (for example, "scan Metasploitable 2 with Nmap and list all open ports and their services").

After completing each practical exercise, I documented my steps, commands used, and findings in a lab notebook, and these were reviewed by the supervisor in a short follow-up discussion. Feedback from these reviews — for instance, being asked to also record service versions alongside open ports, or to cross-check a finding against the Exploit Database — was incorporated into subsequent exercises, which helped refine both my technical accuracy and my habit of documenting work in a way that could be independently verified.

Chapter 5: Bibliography / References

1. Offensive Security. Kali Linux Documentation. Retrieved September 2026, from <https://www.kali.org/docs/>

2. Nmap Project. Nmap Reference Guide. Retrieved September 2026, from <https://nmap.org/book/man.html>
3. VirusTotal. VirusTotal — Analyse suspicious files, domains, IPs and URLs. Retrieved September 2026, from <https://www.virustotal.com/>
4. Offensive Security. Exploit Database. Retrieved September 2026, from <https://www.exploit-db.com/>
5. Shodan. Shodan — Search Engine for the Internet of Everything. Retrieved September 2026, from <https://www.shodan.io/>
6. OverTheWire. Bandit Wargame. Retrieved September 2026, from <https://overthewire.org/wargames/bandit/>
7. Academy of Skill Development. Internship Training Materials and Practical Exercise Guidance, 2026.